Kubernetes as the Platform for AI Infrastructure

Adoption, core technologies, vendor offerings and production experience with training and inference on Kubernetes, 2025–2026

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Abstract

As of September 2026, Kubernetes is the default control plane for AI inference. For training it is a growing platform, but it still depends on add-ons. In the CNCF 2025 Annual Cloud Native Survey, released in January 2026, 66% of organisations hosting generative AI models use Kubernetes for some or all of their inference, and 82% of container users run Kubernetes in production [1,2]. No comparable official figure exists for training on Kubernetes. Over half of organisations do not train models at all [1,3,4].

Adoption is broad but shallow. In the same survey, 44% of organisations do not yet run AI/ML workloads on Kubernetes, and only 7% deploy models daily [1,3]. The economics favour Kubernetes' strengths anyway. Gartner forecasts that inference will account for 55% ($23.3B) of AI-optimised IaaS spending in 2026, overtaking training for the first time, and 59% in 2027 [5].

The inference stack has converged on shared upstream parts. These are the stable v1 InferencePool API with a model-aware Endpoint Picker [6,7,8], llm-d (a CNCF Sandbox project since March 2026) [9,10,11,12], and KServe's LLMInferenceService built on llm-d [13]. NVIDIA Dynamo with Grove is the main parallel stack [14,15]. KV-cache-aware routing is the best-evidenced performance lever. Google reports a 35% lower time to first token for one Vertex AI model and a 52% better P95 for another [16]. These figures, like all others in this area, are self-reported.

Accelerator management now rests on Dynamic Resource Allocation (DRA). Its core APIs went GA in v1.34 (September 2025) [17], and GPU partitioning, consumable capacity and device health moved to beta in v1.36 (April 2026) [18,19]. Training at scale is less mature. Native workload-aware (gang) scheduling is still alpha in v1.36 [19,20], and GPU allocation in NVIDIA's DRA driver is experimental [21]. Large training clusters therefore rely on Kueue, Volcano, KAI Scheduler or Slurm running on Kubernetes [20,21,22,23,24,25].

Vendors build on the same primitives, checked through CNCF's Kubernetes AI Conformance programme. It launched in November 2025 and had 31 certified platforms by March 2026 [26,27]. Vendors compete on control-plane scale, managed operations and vertical integration [16,24,25,26,28,29,30,31,32]. For platform teams, the practical priorities are the ones production users say matter most: queueing and GPU sharing to raise utilisation, and faster movement of model weights [33,34,35,36,37].

Highlights

- **Kubernetes is mainly an inference platform for AI today.** — In the CNCF 2025 survey (released 20 January 2026), 66% of organisations hosting generative AI models run some

or all inference on Kubernetes [1,2]. No official training-on-Kubernetes figure exists, and over half of organisations do not train models [1,3,4].

- **Adoption is wide but operationally immature.** — 44% of organisations do not yet run AI/ML on Kubernetes and only 7% deploy models daily [1,3]. A separate 2025 study found 35% of platform teams do not orchestrate AI workloads [38].
- **DRA is the accelerator foundation, but gang scheduling for training is still alpha upstream.** — Core DRA went GA in v1.34, and partitionable devices and device health went beta in v1.36 [17,18,19]. Workload-aware scheduling and PodGroup are alpha in v1.36 [19,20], so training clusters rely on Kueue, Volcano, KAI or Slurm-on-Kubernetes [20,21,22,23].
- **The Kubernetes inference stack has converged on InferencePool, llm-d and LeaderWorkerSet, with NVIDIA Dynamo as the main alternative.** — InferencePool is stable at v1 and its production Endpoint Picker now lives in llm-d [6,7,8]. KServe v0.17 and OpenShift AI 3 build on llm-d [13,32]. Grove is being integrated with llm-d [31,39].
- **KV-cache-aware routing delivers the best-documented gains, all of them self-reported.** — Google reports up to 40% lower TTFT and inter-token latency (ITL) from llm-d predicted-latency scheduling in Vertex AI [40]. Baseten reports 48–49% lower P95/P99 latency with Dynamo's router [41]. Tesla reports 3x output tokens/s on one deployment [42,43].
- **Utilisation and cost gains come from scheduling and sharing more than from hardware.** — China Merchants Bank pooled nearly 10,000 accelerators under Kueue, KEDA, HAMi and Fluid [33]. It raised utilisation from 35% to over 60% and cut inference cost per million tokens by over 60% [33].
- **At the frontier, power and networking limit scale more than Kubernetes does.** — Google ran a 130,000-node experimental GKE cluster and named electrical power as the main constraint (about 2,700 W per GB200 GPU) [28]. EKS has supported 100,000 nodes since July 2025 [29].

1 Why AI workloads are moving onto Kubernetes

The strongest adoption signal is about inference. The CNCF 2025 Annual Cloud Native Survey was released on 20 January 2026. It found that 66% of organisations hosting generative AI models use Kubernetes for some or all of their inference workloads, and that 82% of container users run Kubernetes in production, up from 66% in 2023 [1,2]. Official materials give only the combined 66% figure. We could not confirm a reported split into 'all' versus 'some' inference in any source [1,3,4,44].

Beneath that headline, adoption is shallow. Only 7% of organisations deploy AI models daily, 47% deploy them occasionally, and 44% do not yet run AI/ML workloads on Kubernetes at all [1,3]. A CNCF blog from August 2026 cites 2025 platform-engineering research finding that 35% of platform teams still do not orchestrate AI workloads [38].

Table 1: Key figures from the CNCF 2025 Annual Cloud Native Survey

Metric	Value	Note
Container users running Kubernetes in production	82%	Up from 66% in 2023
GenAI-hosting organisations using Kubernetes for some or all inference	66%	No official all/some split
Organisations not yet running AI/ML on Kubernetes	44%	Official announcement

Metric	Value	Note
Organisations deploying AI models daily	7%	47% deploy occasionally
Organisations that do not train models	Over half	Linux Foundation blog

Source: [1], [2], [3], [4], [44]. Data as of 2025 survey, published 2026-01-20. Secondary coverage reports a sample of 628 respondents. The official CNCF and Linux Foundation pages do not state the sample size [1,4,45,46].

NO TRAINING-ADOPTION FIGURE

No official CNCF or Linux Foundation page gives a share of organisations training models on Kubernetes. The 36% 'training' figure in the press release is a skills challenge, not model training [1,3,4]. Public survey data covers inference far better than training [1,3,47].

1.1 Spending is shifting to inference

Gartner (August 2026) forecasts worldwide AI-optimised IaaS spending to grow 96% to about $42B in 2026. Of that, $23.3B (55%) is forecast for inference and $19B for training. Gartner expects the inference share to reach 59% in 2027, in a market of about $66B [5]. Deloitte measures compute rather than spending. It estimates inference at about one-third of AI compute in 2023 and about half in 2025, and projects about two-thirds in 2026 [48].

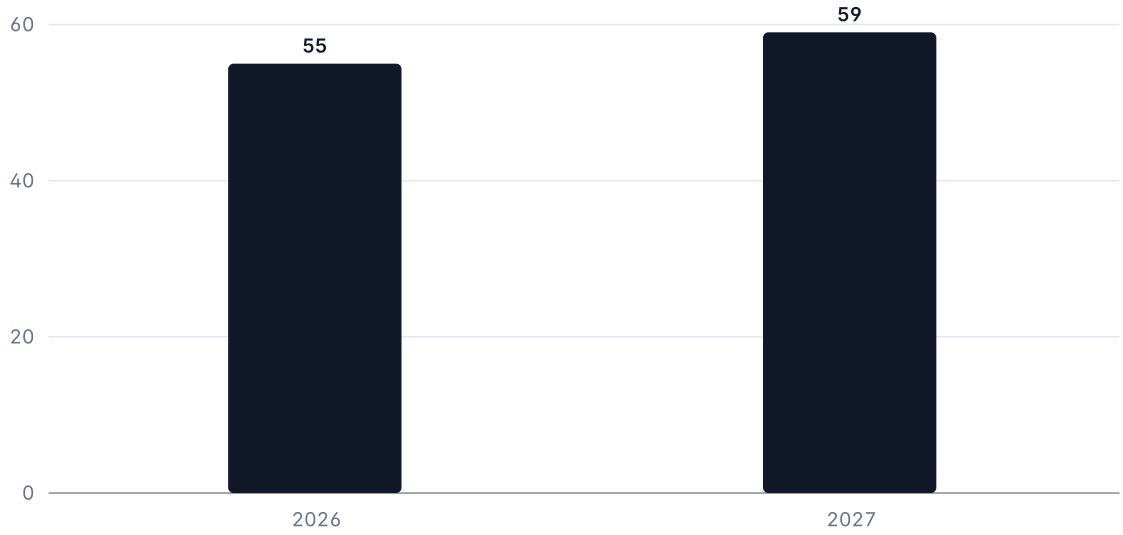


Figure 1: Inference share of AI-optimised IaaS spending (Gartner forecast) (% of spending)

Source: [5]. Data as of Forecast published 2026-08-10. Estimates, not measured spending.

The two forecasts measure different things and neither is a precise market share, but they point the same way. This suggests that the elastic, service-style workloads Kubernetes handles best will make up a growing share of AI infrastructure [5,48].

2 Core building blocks: accelerator management with DRA

Dynamic Resource Allocation replaces the fixed-count device-plugin model with claimable, parameterised devices. Its core resource.k8s.io APIs reached GA in Kubernetes v1.34 on 1 September 2025 [17]. v1.36 (22 April 2026) moved the features that matter most for GPUs into beta, enabled by default. These are partition-

able devices, consumable capacity, and device taints and tolerations [18,19].

Table 2: DRA and related feature maturity by Kubernetes release

Feature	Status	Release	What it enables
Core DRA APIs (resource.k8s.io)	GA	v1.34 (Sep 2025)	Structured device claims beyond device plugins
AdminAccess for ResourceClaims (KEP-5018)	GA	v1.36 (Apr 2026)	Privileged admin access to in-use devices
Partitionable devices	Beta, on by default	v1.36	GPU slicing (e.g. MIG-style partitions)
Consumable capacity	Beta, on by default	v1.36	Finer-grained device sharing
Device taints and tolerations	Beta	v1.36	Keep specialised devices for the right workloads
ResourceClaim device status	Beta	v1.36	Device status reporting on claims
allocatedResourcesStatus in Pod status (KEP-4680)	Beta	v1.36	Shows Unhealthy/Unknown devices behind crash loops
ResourceClaims for PodGroups (KEP-5729)	Alpha in v1.36; beta in v1.37	v1.36 / v1.37	Claim reserved per group; removes the 256-pod reservedFor limit

Source: [17], [18], [19], [20], [49], [50]. Data as of 2026-05.

Device health reporting closes a long-standing operational blind spot. With allocatedResourcesStatus at beta, operators can see in 'kubectl describe pod' whether a crash loop comes from an unhealthy device. This works for both device-plugin and DRA devices [19].

2.1 NVIDIA's DRA driver and multi-node NVLink

NVIDIA's DRA driver provides ComputeDomains, an officially supported abstraction for secure Multi-Node NVLink on systems such as GB200. It sets up IMEX channels and ties the domain's lifetime to the workload [21]. The driver's GPU-allocation component is still experimental and off by default [21]. At KubeCon Europe in March 2026, NVIDIA donated the driver to community governance under the Kubernetes project [31,39].

> **CAUTION**
>
> DRA is GA as an API, but the most common use, allocating NVIDIA GPUs through DRA, is not yet officially supported in NVIDIA's driver [21]. Most GPU clusters still depend on device plugins or vendor tooling for day-to-day allocation. This last point is our inference.

WG Device Management set its 2026 priorities in June. They are multi-node allocation through integration with workload-aware scheduling, device health and failure detection, advanced sharing, and portable specifications [51].

3 Training on Kubernetes: gang scheduling, operators and Slurm

Distributed training needs all-or-nothing placement: a job whose workers cannot all start wastes every GPU it holds. Kubernetes first added native gang scheduling as alpha in v1.35 [20,50,52,53]. v1.36 reworked it into Workload Aware Scheduling, with a static Workload API, a decoupled runtime PodGroup API integrated with the Job controller, and a scheduling cycle that evaluates each group atomically. All of it is still alpha [19,20].

Production training therefore runs on add-on schedulers. Kueue's queueing, quotas with fair sharing and preemption, timeout-based all-or-nothing admission and topology-aware scheduling are GA. MultiKueue multi-cluster dispatch reached GA in February 2026 [54,55]. Kueue integrates with Job, JobSet, Kubeflow jobs, Ray and LeaderWorkerSet, and its latest stable release is v0.19.5 from 17 September 2026 [54,55]. Volcano v1.14 added gang scheduling at PodGroup and SubGroup level with network-topology-aware bin-packing. v1.15 (30 May 2026) added gang-level preemption and brought DRA devices under queue quotas [56,57].

NVIDIA's KAI Scheduler, the open-sourced Run:ai scheduler, joined the CNCF Sandbox in March 2026 [31,39]. It provides the hierarchical gang scheduling that Grove relies on [14,15]. On the operator side, Kubeflow Trainer v2.2 (20 March 2026) merges the old per-framework job types into one TrainJob API on JobSet. It lists DRA, multi-node NVLink and native workload-aware scheduling as roadmap items [23]. Managed platforms fill the same gap. SageMaker HyperPod on EKS added gang scheduling with timeout-based requeueing in April 2026 [58].

Table 3: Training schedulers and Slurm-on-Kubernetes options

Option	Gang scheduling	Topology awareness	Multi-cluster	Maturity signal
Kubernetes WAS / PodGroup	Native, atomic group cycle	Included in WAS work	No	Alpha in v1.36
Kueue	Timeout-based all-or-nothing admission (GA)	Topology-aware scheduling (GA)	MultiKueue (GA Feb 2026)	v0.19.5, Sep 2026
Volcano	PodGroup and SubGroup gangs; gang-level preemption	HyperNode (switch/rack) bin-packing	Not documented in corpus	v1.15, May 2026
KAI Scheduler	Hierarchical gang scheduling (used by Grove)	Topology-aware placement	Not documented in corpus	CNCF Sandbox, Mar 2026
Slurm on Kubernetes (SUNK, Slinky)	Slurm semantics	Topology-aware scheduling (SUNK); dynamic topology (Slinky v1.1.0)	Not documented in corpus	Production at CoreWeave and NVIDIA

Source: [14], [15], [19], [20], [24], [25], [31], [39], [54], [55], [56], [57]. Data as of 2026-09.

3.1 Slurm and Kubernetes are converging

CoreWeave's SUNK runs Slurm login and compute nodes as pods on its Kubernetes service. A syncer and a Slurm-aware pod scheduler let batch jobs and Kubernetes services share one compute pool, with GPU straggler detection added [24]. NVIDIA reports running SchedMD's Slinky slurm-operator across internal clusters totalling more than 8,000 GPUs. It measured NCCL collective performance matching non-containerised Slurm, with a limit of one worker pod per node [25]. NVIDIA announced its acquisition of SchedMD on 15 December 2025 and committed to keeping Slurm open source [59].

INSIGHT

Teams with HPC-style training habits do not have to choose between Slurm and Kubernetes. The evidence points to Slurm semantics running on a Kubernetes substrate that is shared with inference [24,25]. This is our interpretation.

4 Inference on Kubernetes: model-aware routing, disaggregated serving and autoscaling

The Gateway API Inference Extension is the routing layer. Its InferencePool API is stable at v1 and groups model-server Pods behind a standard Gateway such as Envoy Gateway, kgateway, GKE Gateway, NGINX or Istio. An Endpoint Picker (EPP) chooses a replica for each request from its KV-cache utilisation, queue length and active LoRA adapters [6,7,8]. In v1.6.0 the production EPP moved into the llm-d repository, and a lightweight reference EPP remains for conformance [6,7,8].

llm-d supplies the serving stack: vLLM plus the inference scheduler, tiered KV-cache offload, prefill/decode disaggregation and wide expert parallelism. It joined the CNCF Sandbox in March 2026 [9,10,11,12]. Its v0.9 release on 28 August 2026 added router high availability, KEDA autoscaling on KV-cache pressure and queue depth, and revision-aware routing for disaggregated deployments [9,10,11,12]. The same release added support for GB200/GB300, Intel XPU, ROCm and TPU v7 [9,10,11,12].

Higher-level platforms wrap these parts. KServe v0.17 (13 March 2026) made LLMInferenceService production-ready on llm-d. It adds KV-cache-aware routing via the EPP, prefill/decode disaggregation, and multi-node tensor, data and expert parallelism via LeaderWorkerSet. Its Workload Variant Autoscaler scales on KV-cache use and queue depth [13]. LeaderWorkerSet deploys a leader and its workers as one replica. Its DisaggregatedSet API, co-designed with llm-d, coordinates prefill/decode rollouts, though its own gang scheduling is still alpha [22].

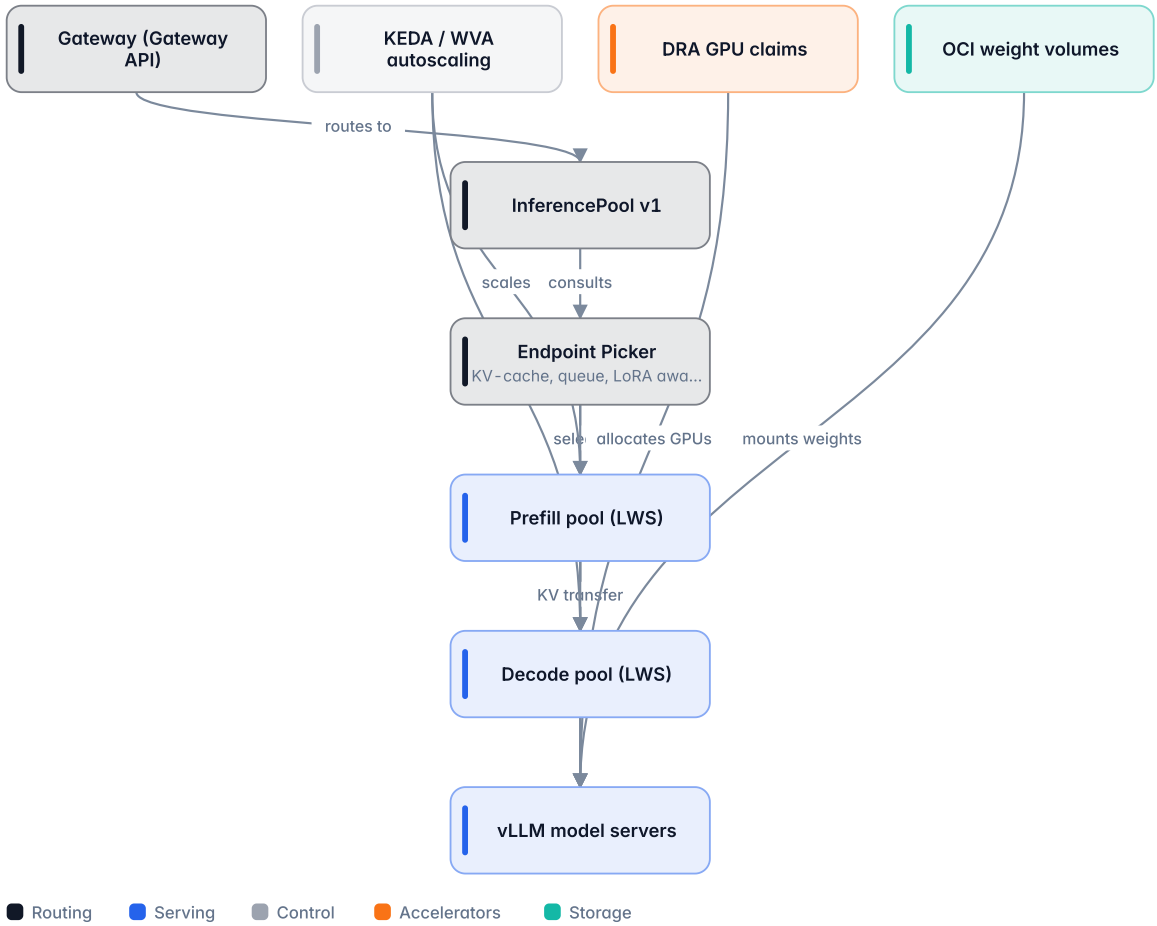


Figure 2: Converged Kubernetes LLM serving stack

Source: [6], [7], [8], [9], [10], [11], [12], [13], [18], [19], [22].

4.1 NVIDIA Dynamo and Grove

NVIDIA Dynamo models an inference graph as a custom resource managed by an operator. Grove orchestrates the router, prefill and decode roles as PodCliques inside a PodCliqueSet, and uses KAI Scheduler for hierarchical gang scheduling [14,15]. Dynamo 1.0 shipped on 16 March 2026 with 17 named production adopters and integrations with managed services including EKS and AKS [60]. NVIDIA says Grove's API is being integrated with llm-d, which should narrow the gap between the two stacks [31,39].

4.2 Model weights and cold starts

Getting weights onto GPUs quickly is the main cold-start problem. OCI Artifact and Image volume sources (KEP-4639) went stable in v1.36, so weights can be packaged as OCI artifacts and mounted from a registry [19]. On GKE, the Cloud Storage FUSE serving profile cut load time for a 480 GB Qwen3-235B-A22B model on TPUs from 39 hours to 14 minutes, according to Google (April 2026) [61].

> **CAUTION**
>
> No independent, like-for-like benchmark compares llm-d, KServe, Dynamo and other stacks. Every performance figure in this report comes from a vendor or adopter.

5 Vendor and managed platform landscape

The major platforms package the same upstream primitives and differentiate on three things: control-plane scale, managed operations and vertical integration [16,24,25,26,28,29,30,31,32]. The first certified platforms in CNCF's AI Conformance programme (November 2025) were EKS, GKE, Microsoft Azure, Red Hat OpenShift and Broadcom VKS, with CoreWeave, Oracle Cloud and Akamai among the supporters [26].

Table 4: How major vendors package AI on Kubernetes

Vendor	Cluster scale	Inference layer	Training / scheduling layer	Notable differentiator
Google GKE	65,000 nodes supported; 130,000 in an experiment (Nov 2025)	GKE Inference Gateway (prefix-cache and load-aware)	Kueue; TPU dynamic slicing	Spanner-based control-plane store; GCS FUSE serving profile
Amazon EKS / SageMaker HyperPod	100,000 nodes (since Jul 2025)	Dynamo integration; gateway details not researched	HyperPod task governance with gang scheduling (Apr 2026)	Reworked etcd storage layer
Microsoft AKS	Not stated in corpus	KAITO (vLLM), AI Runway, Dynamo-Grove on AKS	AKS on Bare Metal (preview), Anyscale managed Ray (preview)	Fleet Manager GA for Arc-enabled clusters
Red Hat OpenShift AI 3	Not stated in corpus	llm-d GA (NVIDIA and AMD)	Not covered in corpus	Model-as-a-Service, AI Hub, agent API
NVIDIA	N/A (software)	Dynamo 1.0, Grove	KAI Scheduler, DRA driver, Slurm via SchedMD/Slinky	Vertical integration across the stack
CoreWeave	Not stated in corpus	llm-d founding contributor	SUNK (Slurm on CKS)	Slurm and Kubernetes share one compute pool

Source: [9], [10], [11], [12], [14], [15], [16], [24], [25], [28], [29], [30], [31], [32], [39], [58], [59], [60], [61], [62]. Data as of 2026-09.

Scale engineering is where the hyperscalers diverge most visibly. Google's 130,000-node experiment replaced etcd with a Spanner-based store and handled 13,000 QPS of lease updates. It created 130,000 Pods in 3 minutes 40 seconds and preempted 39,000 Pods in 93 seconds [28]. AWS rebuilt EKS's etcd storage layer to support 100,000 nodes, which it equates to up to 800,000 NVIDIA GPUs [29].

Microsoft's Build 2026 announcements follow a strategy of wrapping open source (Kubernetes, Ray, Gateway API) in managed services [30]. Red Hat made llm-d generally available in OpenShift AI 3 in October 2025 [32]. NVIDIA is the most vertically integrated. It spans GPU allocation (DRA driver), scheduling (KAI), serving (Dynamo, Grove) and, through SchedMD, Slurm [31,39,59].

> **NOTE**
>
> The corpus contains no cost comparison between managed offerings, and no detail on the EKS and AKS inference gateways. Treat the table as a capability map, not a buyer's ranking.

6 Production case studies

6.1 China Merchants Bank: one pool for training and inference

Context and problem: training needs stable reserved capacity while inference needs fast elastic scaling. China Merchants Bank had nearly 10,000 heterogeneous accelerator cards to serve both [33].

Approach: the bank built one Kubernetes control plane. It uses Kueue for admission and quotas, KEDA and Prometheus for inference autoscaling, HAMi for fine-grained accelerator sharing and Fluid for faster access to data and weights, with 99% of accelerator resources under the unified framework [33].

Results: average accelerator utilisation rose from 35% to over 60%, and inference cost per million tokens fell by more than 60% [33]. Five LoRA fine-tuning tenants sharing one base-model instance cut accelerator use by 80% [33].

Lessons: in this case the gains came from queueing, sharing and pooling, not from new hardware. This is consistent with other deployments, where default Kubernetes scheduling proved insufficient for shared GPUs [33,34,35,36,37].

6.2 Other named deployments

Table 5: Named AI-on-Kubernetes deployments and reported outcomes

Organisation	Workload	Stack	Reported outcome	Source / date
Google Vertex AI	LLM inference	GKE Inference Gateway; llm-d predicted-latency scheduling	TTFT −35% (Qwen3-Coder); P95 TTFT −52% (DeepSeek V3.1); up to 40% lower TTFT/ITL	Google, Feb and Mar 2026
Tesla	Hundreds of LLMs	KServe LLMInferenceService, llm-d, vLLM, Envoy, Inference Extension	3x output tokens/s, 2x lower TTFT on one deployment (Llama 3.1 70B, 4x MI300X)	KServe/llm-d blog, Apr 2026
Baseten	LLM inference (Qwen3 Coder 480B)	NVIDIA Dynamo KV-aware router	P95 −48%, P99 −49% latency on production traffic	Baseten, Mar 2026
Wayve	Multi-tenant inference, evaluation, synthetic data	Kueue, Ray	Predictable allocations and better utilisation (qualitative)	KubeCon EU keynote, Mar 2026
Subaru	Driver-assistance AI development	On-prem Kubernetes, Gateway API, Harbor, Argo	30 GB+ image pulls from ~3 hours to ~3 minutes	CNCF, Jul 2026
Anthropic, Amazon Nova	Ultra-scale workloads on EKS	EKS 100K-node clusters	Anthropic: writes within 15 ms rose from 35% to 90%	AWS, Jul 2025 (historical)
Uber	Michelangelo ML training	Ray on Kubernetes, federated scheduler	1.5–4x faster training	Uber, Apr 2025 (historical)

Source: [16], [29], [34], [35], [37], [40], [41], [42], [43]. Data as of 2026-09.

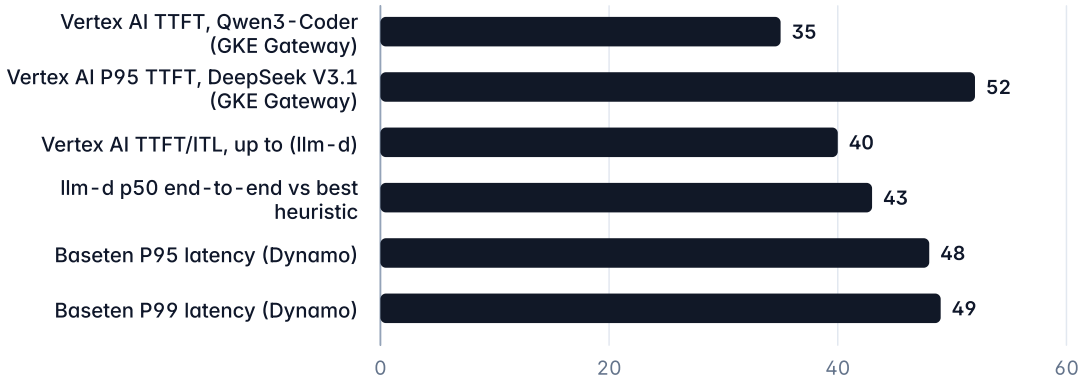


Figure 3: Self-reported latency reductions from model-aware routing (% reduction)

Source: [16], [40], [41]. Data as of 2026-02 to 2026-03. Different models, workloads and baselines. Not directly comparable.

WEAKER-SOURCED FIGURES

HAMi's KubeCon China recap reports that ICBC's GPU utilisation rose from 20% to 70% and that SF Express cut its GPUs from 1,400 to 1,000. These are self-reported in a project recap [63]. The Character.AI figures (60% faster model loading, 50% lower TCO) and Anthropic's TPU dynamic-slicing figures trace only to conference videos [36,62,64,65,66]. Google's official documentation states dynamic slicing gives up to 5x faster startup and 4.5x faster recovery [62].

Earlier milestones are historical. OpenAI reported scaling one Kubernetes cluster to 7,500 nodes in January 2021, per a secondary summary [67]. In November 2024, Anthropic's Head of Compute endorsed GKE's 65,000-node support as 'the scale we need' [68]. That quote does not by itself establish which workloads Anthropic runs on GKE.

7 Limitations, operational challenges and open problems

- Gang scheduling is not native yet. Workload Aware Scheduling and PodGroup are alpha in v1.36, and LeaderWorkerSet gang scheduling is alpha [19,20,22]. Training clusters depend on Kueue, Volcano, KAI or managed equivalents [20,21,22,23].
- GPU allocation through DRA is not production-ready from NVIDIA. The driver's GPU-allocation path is experimental and off by default, and dynamic MIG is still planned [21]. Kubeflow Trainer lists DRA and multi-node NVLink as future work [23].
- Utilisation and multi-tenancy are the main cost levers. Default scheduling falls short for shared GPUs [35], and China Merchants Bank started from 35% average utilisation [33]. The corpus has no tier-1 comparison of the isolation trade-offs between hardware partitioning and software sharing such as HAMi.
- Moving weights and images is a bottleneck. Tesla cited storage problems with multi-hundred-GB models and manual recovery from local-PV affinity [42,43]. Subaru's 30 GB+ images took about 3 hours to pull before its platform rebuild [34].
- At the frontier, the limits are power, networking and multi-cluster operation. Google names electrical power as the main limit on scale and lists managed DRANET for RDMA and MultiKueue as future work [28].
- Evidence on failure handling and goodput is thin. Beyond NVIDIA's finding that Slinky adds no measurable NCCL overhead [25], we found no quantitative comparison of large-run resilience on Kubernetes versus bare-metal Slurm.

NOTE

Non-NVIDIA accelerators are under-documented. Apart from release notes, such as llm-d v0.9 support for ROCm, Intel XPU and TPU v7 [9,10,11,12], the corpus has one AMD production data point (Tesla on MI300X) [42,43].

8 Standards and community direction

CNCF's Certified Kubernetes AI Conformance programme launched on 11 November 2025. It covers GPU resource management, distributed scheduling, accelerator autoscaling and monitoring [26]. By 24 March 2026, 31 platforms were certified, up from 18. The v1.35 requirements added high-performance pod-to-pod communication, advanced inference ingress, disaggregated inference and agentic workloads [27]. The conformance repository accepts certifications for v1.33 through v1.37, each valid for one year. v1.37 introduced hybrid verification that combines automated tests with manual attestation [69].

Community governance has spread out from a single working group. WG Serving concluded on 26 February 2026 and handed its work to SIG Node, Scheduling, Apps, Network and Scalability and to WG Device Management [70]. A new AI Gateway Working Group, announced on 9 March 2026, builds on the Inference Extension to standardise networking for AI traffic [71].

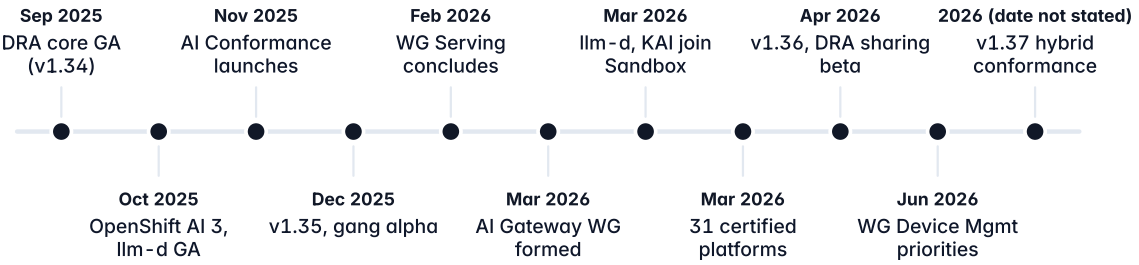


Figure 4: Key milestones for AI on Kubernetes

Source: [9], [10], [11], [12], [17], [18], [19], [20], [26], [27], [31], [32], [39], [50], [51], [52], [53], [69], [70], [71].

9 Outlook and recommendations for platform teams

The recommendations below are our interpretation of the evidence, ordered by how mature the underlying components are.

- Adopt Kueue now for queueing, quotas and all-or-nothing admission. Its core features and MultiKueue are GA [54,55], and it underpins the China Merchants Bank, Wayve and Google 130,000-node results [28,33,35].
- Plan device management around DRA. Use the v1.36 beta sharing and health features [18,19], and keep device plugins or vendor tooling for NVIDIA GPU allocation until the driver's path leaves experimental status [21].
- Standardise inference routing on InferencePool with a production EPP. Choose llm-d/KServe or Dynamo for the serving layer, and design autoscaling around KV-cache and queue signals rather than CPU [6,7,8,9,10,11,12,13].
- Treat native gang scheduling as upcoming, not available. Workload Aware Scheduling is alpha [19,20], and Kueue and Volcano API versions are still changing (Kueue v0.20 stops serving v1beta1) [54,55].

- Budget for weight distribution. Evaluate OCI image volumes, now stable [19], and storage-side acceleration such as GKE's FUSE serving profile [61].
- Require current AI Conformance certification from vendors, and check the Kubernetes version it covers [69].

INSIGHT

The largest reported gains come from pooling and sharing GPUs across tenants and workload types, not from swapping serving frameworks [33,34,35,36,37]. Organisational and scheduling work is likely to repay more than stack selection. This is our inference.

10 Conclusion

For most organisations, the question is no longer whether to run AI inference on Kubernetes. It is how quickly they can move from occasional deployments to operating AI as a routine platform service. The 7% of organisations deploying models daily [1,3] marks how early most teams still are. The forecast shift of spending toward inference [5] means that gap will increasingly show up as cost.

Three developments will decide whether Kubernetes also becomes the default for large-scale training. The first is promotion of Workload Aware Scheduling and PodGroup-scoped DRA claims beyond alpha and beta [19,20,49,50]. The second is official GPU allocation in NVIDIA's community-governed DRA driver [21,31,39]. The third is whether Slurm-on-Kubernetes under NVIDIA's ownership of SchedMD stays vendor-neutral in practice [25,59].

On inference, watch whether Grove's integration with llm-d merges the two leading stacks [31,39], and whether the AI Gateway Working Group standardises traffic handling beyond single-model routing [71]. The evidence also still lacks independent benchmarks. Until neutral, like-for-like results exist, platform teams should reproduce vendor latency claims on their own traffic before committing.

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A Methodology and evidence quality

This paper was produced by Boku, a multi-agent research pipeline. Research workstreams (technical, market, competitive, case-study) gathered evidence from public sources; each claim was recorded with its sources and date. An independent fact-checking agent reviewed the corpus over 2 round(s) (final status: pass) before synthesis and editing. Information dated within 365 days of publication is labelled current; older material is labelled historical.

Table 6: Evidence base

Measure	Count
Findings recorded	72
Findings used (after fact-check)	70
Findings rejected	2
Findings flagged as weakly supported	8
Distinct sources supporting usable findings	75
Tier 1 · Primary / official	56
Tier 2 · Authoritative secondary	7
Tier 3 · Specialist / analyst	9
Tier 4 · Community / unverified	3

Limitations and unresolved questions

- Unverified: F055 p50 TTFT 0.97s / 70% reduction figures in the llm-d predicted-latency blog
- Unverified: F058 Baseten P95/P99 reductions
- Unverified: F024 v1.37 hybrid verification in the k8s-ai-conformance README
- No official figure for the share of organisations running AI training (as opposed to inference) on Kubernetes. The official survey sample size is not stated (628 comes from secondary coverage only).
- No independent, like-for-like benchmark comparing llm-d, KServe, Dynamo and other serving stacks. All performance figures are self-reported by vendors or adopters.
- Thin evidence on multi-node networking (RDMA/InfiniBand, DRANET) in production on Kubernetes beyond Google's mention of managed DRANET as future work.

- Little quantitative data on failure handling and job goodput for large training runs on Kubernetes compared with bare-metal Slurm, beyond NVIDIA's statement that Slinky adds no NCCL overhead.
- Multi-tenancy security and isolation for shared GPUs (e.g. MIG/MPS vs software sharing such as HAMi) is not covered by tier-1 comparative evidence.
- No cost comparison between managed Kubernetes AI offerings, and no per-vendor details on the EKS and AKS inference gateways.

B Evidence register

Each cited claim with its evidence status. Current: dated within the freshness window. Historical: older than the window. Estimated: a published estimate, not a reported figure. Derived: calculated from reported figures. Inference: analytical interpretation. Unknown: no reliable date.

Table 7: Cited findings

ID	Claim	Status	As of	Sources
F001	Dynamic Resource Allocation (DRA) reached General Availability in Kubernetes v1.34, with the core resource.k8s.io APIs stable. Other DRA features in that release were still beta or alpha.	Historical	2025-09-01	[17]
F003	NVIDIA's DRA driver for GPUs (kubernetes-sigs/dra-driver-nvidia-gpu) provides ComputeDomains, an officially supported abstraction for secure Multi-Node NVLink (e.g. GB200) whose lifetime is tied to the workload. Its GPU-…	Current	2026-09	[21]
F004	Kubernetes has started building native workload-aware and gang scheduling (all-or-nothing placement of a group of Pods). It arrived as alpha (off by default) in v1.35 via the Workload API/PodGroup. In v1.36 it was rework…	Current	2026-05-13	[52,53,20,50]
F005	In Kueue's feature overview, queueing, quotas (ClusterQueues, cohorts, fair sharing, preemption), timeout-based all-or-nothing admission and topology-aware scheduling are GA, and MultiKueue (multi-cluster dispatch) is GA…	Current	2026-09-17	[54,55]
F006	Kubeflow Trainer v2.2 (20 March 2026) replaces separate job types (PyTorchJob, MPIJob, JAXJob, XGBoostJob) with one TrainJob API built on JobSet and reusable TrainingRuntimes, and adds Flux for HPC/MPI. Multi-node NVLink…	Current	2026-03-20	[23]
F007	Volcano v1.14 added gang scheduling at both PodGroup and SubGroup level and network-topology-aware bin-packing across HyperNodes (switch/rack). Volcano v1.15 (30 May 2026) added gang-level preemption and put DRA devices …	Current	2026-05-30	[56,57]
F008	LeaderWorkerSet (LWS) runs a leader Pod plus workers as one unit that is replicated, rolled out and placed together. It is the standard primitive for serving a model sharded across multiple hosts. Its all-or-nothing gang…	Current	2026-09	[22]

ID	Claim	Status	As of	Sources
F009	The Gateway API Inference Extension is an official Kubernetes project with a stable v1 InferencePool API: a Service-like group of model-server Pods behind a Gateway (Envoy Gateway, kgateway, GKE Gateway, NGINX, Istio and…	Current	2026-09	[6,7,8]
F010	llm-d is a Kubernetes-native distributed inference stack (vLLM plus an inference scheduler/EPP, tiered KV-cache and LWS-based prefill/decode and wide expert-parallelism). It joined the CNCF as a Sandbox project in March …	Current	2026-08-28	[9,10,11,12]
F011	KServe v0.17 (13 March 2026) made LLMInferenceService production-ready. It is built on llm-d and provides KV-cache-aware routing through the Inference Extension's EPP, disaggregated prefill/decode, multi-node TP/DP/EP vi…	Current	2026-03-13	[13]
F012	On Kubernetes, NVIDIA Dynamo describes inference graphs as custom resources (e.g. DynamoGraphDeployment), managed by an operator. Grove handles orchestration: PodCliqueSet contains PodCliques (router, prefill, decode) an…	Current	2026-06-02	[14,15]
F013	In Nov 2025 Google reported running a 130,000-node GKE cluster in experimental mode, double the officially supported 65,000. It used a Spanner-based store in place of etcd, consistent reads from cache (KEP-2340), a snaps…	Current	2025-11	[28]
F015	Several gaps remain at the core-Kubernetes level: native gang/workload scheduling is still alpha, LWS gang scheduling is alpha, GPU allocation in NVIDIA's DRA driver is experimental, and Kubeflow Trainer lists DRA and MN…	Inference	2026-09	[20,22,21,23]
F016	The CNCF 2025 Annual Cloud Native Survey, released 20 January 2026, found that 66% of organisations hosting generative AI models use Kubernetes for some or all of their inference workloads, and that 82% of container user…	Current	2025	[1,2]
F018	The CNCF 2025 survey found that only 7% of organisations deploy AI models daily and 47% deploy occasionally, that 44% do not yet run AI/ML workloads on Kubernetes, and that over half do not train models at all.	Current	2025	[1,3]
F019	A CNCF blog post from August 2026 cites the '2025 State of AI in Platform Engineering' research, which found that 35% of platform teams still do not orchestrate AI workloads.	Current	2025	[38]
F020	Deloitte (TMT Predictions 2026, published 18 Nov 2025) says inference was about one-third of all AI compute in 2023 and about half in 2025, and projects about two-thirds in 2026.	Estimated	2026	[48]

ID	Claim	Status	As of	Sources
F023	By 24 March 2026 (KubeCon Europe, Amsterdam), CNCF reported 31 certified Kubernetes AI platforms, up from 18. New entrants included OVHcloud, Spectro Cloud, JD Cloud and China Unicom Cloud. The Kubernetes v1.35 requireme…	Current	2026-03-24	[27]
F024	As of September 2026, the CNCF k8s-ai-conformance repository accepts certifications for Kubernetes v1.33 through v1.37. Certifications are valid for one year, and v1.37 introduced hybrid verification that combines automa…	Current	2026-09	[69]
F025	Kubernetes WG Serving concluded on 26 February 2026, having met its goal of making Kubernetes the orchestration platform of choice for inference. Its remaining work moved to SIG Node, SIG Scheduling, SIG Apps (LeaderWork…	Current	2026-02-26	[70]
F026	Kubernetes announced a new AI Gateway Working Group on 9 March 2026 (repo kubernetes-sigs/wg-ai-gateway). It develops standards for gateway and networking infrastructure for AI traffic, building on the Gateway API Infere…	Current	2026-03-09	[71]
F027	WG Device Management, whose core project DRA went GA in Kubernetes 1.34, set these priorities in June 2026: more expressive allocation (including multi-node support through integration with workload-aware scheduling), da…	Current	2026-06-24	[51]
F028	Public survey data covers inference on Kubernetes much better than training on Kubernetes. The CNCF headline metric is inference-specific, and most organisations consume pre-trained models rather than training their own.	Inference	2026-01	[1,3,47]
F029	Google Cloud officially supports 65,000-node GKE clusters. In an experiment published 22 Nov 2025 it ran a 130,000-node cluster, using a Spanner-based storage backend in place of etcd and Kueue for all-or-nothing job que…	Current	2025-11-22	[28]
F030	Since July 2025, Amazon EKS has supported up to 100,000 worker nodes per cluster, which AWS says equals up to 1.6M Trainium chips or 800,000 NVIDIA GPUs. AWS names Anthropic (Claude) and Amazon's Nova team as ultra-scale…	Historical	2025-07-16	[29]
F031	In April 2026, SageMaker HyperPod task governance on EKS-orchestrated clusters added gang scheduling: if not all pods of a distributed training job are ready within a set timeout, the workload is pulled back and requeued…	Current	2026-04	[58]
F032	GKE Inference Gateway extends the standard Kubernetes Gateway API with load-aware and prefix-cache-aware routing. Google reports that Vertex AI's time to first token (TTFT) fell 35% for Qwen3-Coder, P95 TTFT improved 52%…	Current	2026-02-07	[16]

ID	Claim	Status	As of	Sources
F033	At Build 2026, Microsoft announced AKS on Bare Metal (public preview, for NVLink/RDMA training), Anyscale on Azure (managed Ray, public preview), AI Runway (Kubernetes-native model deployment), KAITO for provisioning vLL…	Current	2026-06-23	[30]
F034	At KubeCon Europe in March 2026, NVIDIA donated its DRA driver for GPUs to the CNCF/Kubernetes project, announced that KAI Scheduler (the open-sourced Run:ai scheduler) had joined the CNCF Sandbox, released Dynamo 1.0, a…	Current	2026-03-24	[31,39]
F035	Red Hat OpenShift AI 3.0, announced on 14 Oct 2025 as part of Red Hat AI 3, made llm-d generally available for distributed, disaggregated vLLM inference on NVIDIA and AMD accelerators. It also added Model-as-a-Service, A…	Current	2025-10-14	[32]
F036	CoreWeave's SUNK runs Slurm login and compute nodes as pods on CoreWeave Kubernetes Service (CKS). A syncer and a Slurm-aware pod scheduler let Slurm batch jobs and Kubernetes services share one compute pool. It adds Pyx…	Current	2026-09	[24]
F037	NVIDIA says it runs SchedMD's Slinky slurm-operator in production across several internal clusters totalling more than 8,000 GPUs. It reports that NCCL all-reduce/all-gather performance matches non-containerised Slurm, w…	Current	2026-04-09	[25]
F038	On 15 Dec 2025, NVIDIA announced it would acquire SchedMD, Slurm's developer. It committed to keep Slurm open-source and vendor-neutral, and said Slurm runs on more than half of the TOP500's top-10 and top-100 systems.	Current	2025-12-15	[59]
F039	CNCF's Certified Kubernetes AI Conformance programme launched on 11 Nov 2025. Amazon EKS, GKE, Microsoft Azure, Red Hat OpenShift and Broadcom VKS were among the first certified platforms, with CoreWeave, OCI, Akamai and…	Current	2025-11-11	[26]
F040	Vendors are converging on shared upstream primitives (Gateway API Inference Extension, DRA, Kueue/JobSet, llm-d/vLLM, AI Conformance). They differentiate on control-plane scale engineering (GKE's Spanner backend, EKS's r…	Inference	2026-09	[28,29,16,30,31,32,24,25,26]
F041	China Merchants Bank runs AI training, fine-tuning and online inference on one shared Kubernetes control plane covering nearly 10,000 mixed accelerator cards. The stack is Kueue for admission and quotas, KEDA plus Promet…	Current	2026-09	[33]
F042	According to CNCF, China Merchants Bank raised average accelerator utilisation from 35% to more than 60% and cut inference cost per 1 million tokens by more than 60%. Letting 5 LoRA fine-tuning tenants share one base-mod…	Current	2026-09	[33]

ID	Claim	Status	As of	Sources
F043	At KubeCon China in September 2026, HAMi's recap reported GPU-sharing results from several Chinese users. Intsig runs about 10,000 GPUs with 1,000+ inference services and 1,000+ training jobs, and reports +50% GPU utilis…	Current · weak support	2026-09	[63]
F044	Subaru's EyeSight driver-assistance AI team built an on-prem Kubernetes platform (Envoy Gateway, Gateway API, MetalLB, Harbor, Argo CD, Argo Workflows, Helm/Helmfile). Pull times for AI container images over 30 GB fell f…	Current	2026-07	[34]
F045	Wayve (autonomous driving) runs multi-tenant AI inference on shared GPU Kubernetes clusters. This ranges from latency-sensitive evaluation and validation to large-scale synthetic data generation. Wayve uses Kueue for que…	Current	2026-03-24	[35]
F047	Uber moved all of its Michelangelo ML projects from its legacy Mesos/Peloton-based MADLJ stack to Ray on Kubernetes by early 2024. It uses a federated setup: a global control plane with custom resources and a job schedul…	Historical	2025-04-03	[37]
F048	In November 2024, Google announced that GKE supports clusters of 65,000 nodes (up to 250,000+ accelerators). Anthropic's Head of Compute publicly backed this as the scale Anthropic needs, which points to Anthropic traini…	Historical · weak support	2024-11-14	[68]
F050	OpenAI said in 2021 that it had scaled a single Kubernetes cluster to 7,500 nodes (up from 2,500) for large training jobs such as GPT-3, CLIP and DALL-E. It used one pod per node, gang scheduling, a kube-scheduler tuned …	Historical · weak support	2021-01	[67]
F051	Across the current cases, the same lessons recur. Default Kubernetes scheduling is not enough for shared GPUs, so teams add Kueue, gang scheduling or federated schedulers. GPU sharing and one unified pool for training an…	Inference	2026-09	[33,34,35,36,37]
F052	Gartner forecasts worldwide AI-optimised IaaS spending will grow 96% in 2026 to about $42B. Inference is forecast at $23.3B (55%), overtaking training at $19B, and inference's share is forecast to reach 59% in 2027, with…	Estimated	2026	[5]
F053	At launch (11 Nov 2025), the CNCF Certified Kubernetes AI Conformance programme named AWS EKS, Broadcom VMware vSphere Kubernetes Service, Google Cloud GKE, Microsoft Azure and Red Hat OpenShift as certified. CoreWeave, …	Current	2025-11-11	[26]
F054	Google reports that llm-d's predicted-latency-based scheduling, deployed in Vertex AI production clusters, cut TTFT and inter-token latency by up to 40%.	Current	2026-03-13	[40]

ID	Claim	Status	As of	Sources
F055	On a benchmark built from seven days of real traffic (Qwen3-480B on vLLM, 13 servers with 8x NVIDIA H200 each), predicted-latency routing gave a p50 end-to-end latency of 9.06 s, 43% lower than the best heuristic router.…	Current	2026-03-13	[40]
F056	Tesla needed to run hundreds of LLMs on Kubernetes. It moved from vLLM wrapped in StatefulSets to KServe LLMInferenceService/LLMInferenceConfig with llm-d, vLLM, Envoy and the Gateway API Inference Extension for prefix-c…	Current	2026-04-21	[42,43]
F057	Tesla reported a 3x gain in output tokens/s and a 2x cut in TTFT on one deployment after enabling prefix-cache-aware routing. The deployment served Llama 3.1 70B on 4 AMD MI300X GPUs (tensor parallel 4).	Current	2026-04-21	[42,43]
F058	Baseten reports that NVIDIA Dynamo's KV-cache-aware router, serving Qwen3 Coder 480B, cut P95 latency by 48% and P99 latency by 49% on production OpenRouter traffic. In stress tests it cut average TTFT by 50%, cut TPOT b…	Current	2026-03-16	[41]
F059	At the Dynamo 1.0 launch on 16 Mar 2026, NVIDIA named 17 production adopters: AstraZeneca, Baseten, ByteDance, CoreWeave, Crusoe, DigitalOcean, Gcore, GMI Cloud, Meituan, Nebius, Pinterest, Prime Intellect, Rednote, Soft…	Current	2026-03-16	[60]
F060	The Anthropic TPU dynamic-slicing figures (slice provisioning cut from 10–30 min to 20–60 s, failure recovery from hours to 2–5 min, a 2,048-chip/512-VM slice formed in 41 s) trace back to a video titled 'How Anthropic u…	Current	2026	[65,62,66]
F061	Google's official GKE documentation says dynamic slicing cuts TPU workload startup time by up to 5x and recovery time by up to 4.5x compared with re-creating a whole node pool. It applies to Ironwood (TPU7x). Super-slici…	Current	2026-09-22	[62]
F062	The Character.AI GKE figures (over 20 million users; multiple GKE clusters with separate training and inference node pools; GCS FUSE with local-SSD caching; 60% faster model loading with the Run:ai Model Streamer; Llama …	Current	2026	[36,64]
F063	Google's closest official source on GKE model loading is not Character.AI-specific. On 9 Apr 2026, Google reported that the GKE Cloud Storage FUSE 'gcsfusecsi-serving' profile cut model load time for a 480 GB Qwen3-235B-…	Current	2026-04-09	[61]
F064	The official Kubernetes v1.36 release blog (22 April 2026) says 'DRA: AdminAccess for ResourceClaims and ResourceClaimTemplates' (KEP-5018) graduated to Stable (GA). This gives cluster administrators a permanent, privile…	Current	2026-04-22	[18,19]

ID	Claim	Status	As of	Sources
F065	In Kubernetes v1.36, DRA partitionable devices reached beta and are enabled by default, together with consumable capacity and device taints/tolerations (both beta). ResourceClaim device status also reached beta.	Current	2026-04-22	[18,19]
F066	Kubernetes v1.36 moved the Pod-status field allocatedResourcesStatus (device health reporting, KEP-4680) to beta. Users can now tell from 'kubectl describe pod' whether a crash loop is caused by an Unhealthy or Unknown d…	Current	2026-04-22	[19]
F067	The v1.36 release blog describes Workload Aware Scheduling (WAS) as Alpha. It adds a revised Workload API and a new decoupled PodGroup API, integrated with the Job controller, and a new PodGroup scheduling cycle that eva…	Current	2026-04-22	[19,20]
F068	v1.36 added alpha support for associating DRA ResourceClaims and ResourceClaimTemplates with PodGroups (KEP-5729, 'DRA: ResourceClaim Support for Workloads'). A claim referenced by a PodGroup is reserved for the group as…	Current	2026-05-13	[49,50,20]
F069	The v1.36 release blog confirms that 'VolumeSource: OCI Artifact and/or Image' (KEP-4639) graduated to Stable. This supports packaging model weights as OCI artifacts.	Current	2026-04-22	[19]
F070	The 2025 CNCF Annual Cloud Native Survey covered 628 organisations/respondents with a 65-question questionnaire, according to secondary coverage. The official CNCF press release, the report landing page and the LF resear…	Current	2025-09	[45,1,4,46]
F071	I could not confirm the 23% 'all' / 43% 'some' inference split in any official or secondary source I opened. Official CNCF/LF materials give only the combined figure: 66% of organisations hosting generative AI models use…	Current	2025-09	[1,4,3,44]
F072	No official CNCF/LF page I read gives a figure for running AI training on Kubernetes. The Linux Foundation's blog on the survey says over half of organisations do not train models at all. The official announcement says 4…	Current	2025-09	[3,1,4]

*Compiled from public sources by Boku, an automated multi-agent research pipeline, in response to: “How Kubernetes is being used for AI infrastructure”. This paper reports no new experiments; every result is drawn from the cited work. Verify material facts before relying on them.